Fundamentals of Machine Learning
Final Hard Mock Midterm (Lectures 1 to 5)

Three hours, 100 points, no documents. True/false, multiple choice, then six problems that make you prove something new and judge a situation you have never seen

Fundamentals of Machine Learning

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How this paper is built

Part A (15 pts): ten true/false statements. Each needs a one-line justification: a correct verdict with no reason earns half. **Part B** (16 pts): eight multiple-choice questions, *one or more* options may be correct; 2 points for exactly the right set, 1 point for one wrong or missing option. **Part C** (69 pts): six problems. Each sub-question is tagged **[Prove]** (derive a result the lectures did not derive, from tools they did) or **[Judge]** (diagnose a situation: name the mechanism, say what it does to the number, give the check and the fix). Hints give the one identity you might not remember.

Part A. True or false? Justify in one line. [10 × 1.5 = 15 pts]

A1. If a classifier reaches zero training error on a sample in which no two flows share the same feature vector, the Bayes error of that representation is zero. True / False

A2. Adding a feature to the representation can never increase the Bayes error. True / False

A3. On the same training data, the ridge solution never has a lower training mean squared error than the ordinary least-squares solution. True / False

A4. Two models output probabilities, and one's output is a strictly increasing function of the other's. If each alerts on its 100 highest-scored flows, they flag the same flows, yet their log losses can differ. True / False

A5. Growing a random forest from 100 to 1,000 trees lowers the correlation between its trees. True / False

A6. With squared loss, $F_0 = \bar{y}$ and learning rate $\eta = 1$, one round of gradient boosting with a regression tree gives the same predictions as fitting that tree directly to y. True / False

A7. A feature whose permutation importance is zero carries no information about the label. True / False

A8. After the posterior has been corrected for a new deployment prior, the cost-sensitive threshold $C_{FP}/(C_{FP} + C_{FN})$ must also be recomputed for the new prior. True / False

A9. Gaussian naive Bayes in which each feature has the same variance in both classes has a linear decision boundary. True / False

A10. For any two distributions p and q on the same outcomes, $H(p, q) \geq H(q)$. True / False

Part B. Multiple choice: one or more options are correct. [$8 \times 2 = 16$ pts]

B1. A deep decision tree has training error 0.01 and validation error 0.30. Which changes can be expected to lower the validation error?
(a) more training data from the same distribution (b) a minimum leaf size of 50 (c) adding pairwise interaction features (d) bagging 300 such trees (e) allowing leaves of size 1

B2. A bootstrap sample of size $N = 10$ is drawn from 10 observations. The probability that a given observation appears in it at least once is:
(a) 0.632 (b) 0.651 (c) 0.349 (d) 0.900

B3. Which changes to the score vector s leave $\mathrm{softmax}(s)$ unchanged?
(a) adding the same constant to every score (b) multiplying every score by 2 (c) subtracting $\max_c s_c$ from every score (d) adding a constant to one score only

B4. At some x, $\eta = \mathbb{P}(Y = 1 \mid x) = 0.7$. Which predictions q have a *lower* expected cross-entropy than the hedge $q = 0.5$?
(a) 0.4 (b) 0.6 (c) 0.9 (d) 0.99

B5. A detector's alert threshold is raised. Which quantities can *never increase*?
(a) recall (b) precision (c) the number of alerts (d) the false-positive rate

B6. A node holds 40 attacks and 60 benign flows. A split sends 30 attacks and 10 benign left, 10 attacks and 50 benign right. The decrease in Gini impurity at this node ($I(S) - \frac{|S_L|}{|S|} I(S_L) - \frac{|S_R|}{|S|} I(S_R)$) is closest to:
(a) 0.163 (b) 0.317 (c) 0.480 (d) 0.208

B7. Which hyperparameters appear in the formula XGBoost uses to accept or reject a candidate split?
(a) γ (b) λ (c) the learning rate η (d) the number of rounds

B8. A detector scores each flow at the moment it completes. Which features are legitimate at that moment?
(a) the bytes of this flow (b) the number of flows from the same source in the next 60 s (c) whether the source IP is on a blocklist published the next day (d) the number of distinct ports this source contacted in the previous 60 s

Part C. Problems [69 pts]

C1. Three penalties on one coefficient — 25 minutes — [13 pts]

When the standardised features are orthogonal, penalised least squares separates into one problem per coefficient: minimise $\frac{1}{2}(w - z)^2 + \Omega(w)$, where z is the ordinary least-squares coefficient. The course derived ridge, $\Omega = \lambda w^2 \Rightarrow \hat{w} = z/(1 + 2\lambda)$, and lasso, $\Omega = \lambda|w| \Rightarrow \hat{w} = \mathrm{sign}(z)\max(|z| - \lambda, 0)$.

(a) **[Prove]** Best-subset selection penalises the *number* of non-zero coefficients: $\Omega(w) = \lambda\,\mathbf{1}[w \neq 0]$. Show that $\hat{w} = z$ if $|z| > \sqrt{2\lambda}$ and $\hat{w} = 0$ otherwise (*hard thresholding*). [3 pts]

(b) **[Prove]** The elastic net uses $\Omega(w) = \lambda_1|w| + \lambda_2 w^2$. Show that $\hat{w} = \mathrm{sign}(z)\max(|z| - \lambda_1, 0)/(1 + 2\lambda_2)$. [2 pts]

(c) For $z = (3,\ 0.8,\ -0.4,\ -2)$ compute best subset with $\lambda = 0.5$, lasso with $\lambda = 1$, the elastic net with $\lambda_1 = 1$, $\lambda_2 = 0.5$, and ridge with $\lambda = 0.5$. Which of them leave the large

coefficients untouched, and which shrink them? [3 pts]

(d) **[Prove]** Two columns are identical, $x_2 = x_1$, so the fit depends only on $c = w_1 + w_2$. Show that the lasso objective takes the same value for *every* split of c into w_1, w_2 of the same sign, while the elastic net has the unique minimiser $w_1 = w_2 = c/2$. What does this say about lasso on correlated features? [3 pts]

(e) **[Judge]** Which penalty would you choose (i) to give auditors a short list of the features the detector uses, (ii) with 300 weak, correlated features each carrying a little signal? In general best subset is combinatorial; why did it become trivial in (a)? [2 pts]

Hint. for (d), $|a|+|b| \geq |a+b|$ with equality iff a, b have the same sign, and $a^2+b^2 = \frac{1}{2}(a+b)^2 + \frac{1}{2}(a-b)^2$.

C2. A feature that is the same in both classes — 30 minutes — [14 pts]

A detector watches two servers. x_1 is the standardised log response time of server 1, which slows down by δ during an attack. x_2 is the standardised log response time of server 2, which is never attacked. Both share the same network congestion, so they are correlated:

$$x \mid y = 0 \sim N((0,0)^\top, \Sigma), \qquad x \mid y = 1 \sim N((\delta,0)^\top, \Sigma), \qquad \Sigma = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix},$$

with equal priors, $\delta = 1$, $\rho = 0.8$.

(a) **[Judge]** Show that x_2 has the same distribution in both classes. What does a correlation filter (or any one-feature-at-a-time test) decide about x_2? [2 pts]

(b) **[Prove]** Using $\Sigma^{-1} = \frac{1}{1-\rho^2}\begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}$, derive the LDA weight vector $w = \Sigma^{-1}(\mu_1 - \mu_0)$. Write the score as a function of $x_1 - \rho x_2$ and explain in words what the model does with server 2. [3 pts]

(c) **[Prove]** For two Gaussian classes with a shared covariance and equal priors, the Bayes error is $\Phi(-D/2)$ with $D^2 = (\mu_1 - \mu_0)^\top \Sigma^{-1}(\mu_1 - \mu_0)$. Show that $D^2 = \delta^2/(1-\rho^2)$ and compute the error using both features and using x_1 alone. [3 pts]

(d) **[Judge]** What error does Gaussian naive Bayes reach on (x_1, x_2), and why? Tutorial 1 showed that correlated features make naive Bayes *overconfident*. What is the different failure here? [2 pts]

(e) **[Judge]** Why might a greedy decision tree never split on x_2 at the root, and what does it need to exploit x_2 at all? [2 pts]

(f) **[Judge]** Server 2 is moved to another data centre: at deployment $\rho = 0$, means unchanged, and the fitted LDA model is kept. Show that the separation of its score (difference of the two class means of $w^\top x$, divided by the standard deviation of $w^\top x$) becomes $\delta/\sqrt{1+\rho^2}$, with ρ the training value 0.8, compute the error, and draw the lesson. [2 pts]

Hint. $\Phi(-0.5) = 0.309$, $\Phi(-0.39) = 0.348$, $\Phi(-0.83) = 0.202$.

C3. Bagging the nearest neighbour — 25 minutes — [12 pts]

A 1-NN regressor predicts at a query x_0. The training inputs, sorted by distance to x_0 (no ties), are $x_{(1)}, x_{(2)}, \ldots$ with responses $y_{(i)} = f(x_{(i)}) + \varepsilon_i$, the ε_i independent with variance σ^2; inputs are fixed. Bagging draws bootstrap samples of size N with replacement and averages the 1-NN predictions. Use $(1 - j/N)^N \approx e^{-j}$.

(a) **[Prove]** Show that the probability that $x_{(1)}, \ldots, x_{(j)}$ are *all* absent from a bootstrap sample is $(1 - j/N)^N$. [2 pts]

(b) **[Prove]** Deduce that the probability that $x_{(i)}$ is the nearest point *present* in the sample is $w_i \approx (1-e^{-1})\,e^{-(i-1)}$. Compute $w_1, \dots, w_4$ and check that the w_i sum to 1. [3 pts]

(c) **[Prove]** With infinitely many bootstrap samples the bagged prediction is $\sum_i w_i\, y_{(i)}$. Show that its variance is $\sigma^2 \frac{1-e^{-1}}{1+e^{-1}} \approx 0.46\,\sigma^2$. A k-NN average has variance σ^2/k: which "k" does bagged 1-NN behave like? [3 pts]

(d) **[Judge]** What happens to the bias compared with plain 1-NN, and why? [2 pts]

(e) **[Judge]** A colleague says "bagging only helps trees". Answer with (c). Then explain why bagging a 50-NN regressor changes almost nothing. [2 pts]

Hint. $\sum_{i\ge 0} r^i = 1/(1-r)$ for $|r|<1$, and $1-e^{-2} = (1-e^{-1})(1+e^{-1})$.

C4. Two ways to shrink a boosted tree — 20 minutes — [10 pts]

Squared loss $\ell = \frac{1}{2}(y-F)^2$, $F_0 = \bar{y}$. To isolate the effect of shrinkage, assume every round uses the *same* partition of the inputs into leaves. Consider a leaf L with n_L observations whose mean response is $\bar{y}_L$.

(a) **[Prove]** Show that after m rounds with learning rate η, every point of L is predicted $F_m = \bar{y}_L - (1-\eta)^m(\bar{y}_L - F_0)$. [3 pts]

(b) With $F_0 = 1$, $\bar{y}_L = 5$: compute F_{10} and F_{50} for $\eta = 0.1$, and F_{10} for $\eta = 0.3$. Explain with $(1-\eta)^m \approx e^{-\eta m}$ why a smaller η selects more rounds (Tutorial 2, 7d). [2 pts]

(c) **[Prove]** For this loss, g_i and h_i are the derivatives with respect to F. Show that the XGBoost leaf value is $w^* = \frac{n_L}{n_L+\lambda} \cdot \overline{(y-F)}_L$, a shrunken mean residual. [2 pts]

(d) **[Judge]** Both η and λ shrink the correction. Compare their effect on a leaf of 2 observations and on a leaf of 200, with $\lambda = 10$. Which one is a bias–variance device aimed at small leaves, and why is that sensible? [2 pts]

(e) **[Judge]** In this model, early stopping after m rounds is exactly a form of shrinkage. Toward which value, and by which factor? [1 pts]

C5. How many attacks do we have? — 20 minutes — [10 pts]

A model was trained on a balanced sample ($\pi_t = 0.5$) and is calibrated there. It is deployed on unlabelled traffic.

(a) **[Prove]** Show that for any distribution, $\mathbb{E}_x\big[\mathbb{P}(Y=1 \mid x)\big] = \mathbb{P}(Y=1)$. What monitoring check does this give for a calibrated model on its own distribution? [2 pts]

(b) **[Judge]** On deployment traffic the average output is 0.37. Is the deployment attack rate 0.37? Argue with two extreme models: one that separates the classes perfectly, one that outputs 0.5 for every flow. [2 pts]

(c) **[Prove]** Under prior shift, the corrected posterior is

$$r_d(x) = \frac{(\pi_d/\pi_t)\, r_t(x)}{(\pi_d/\pi_t)\, r_t(x) + \frac{1-\pi_d}{1-\pi_t}\,(1-r_t(x))}.$$

Explain why the true π_d must satisfy $\pi_d = \frac{1}{n}\sum_i r_d(x_i)$. On a toy sample of five flows with $r_t = (0.9,\ 0.6,\ 0.2,\ 0.1,\ 0.05)$, start from $\pi_d = 0.37$ and do one update of this fixed-point equation. [3 pts]

(d) **[Judge]** Name two situations in which this estimate is wrong even with many flows. [1.5 pts]

(e) **[Judge]** Management wants an alarm when the estimated attack rate doubles. Give two changes, not caused by attacks, that would trigger it, and how you would tell them apart from a real increase. [1.5 pts]

C6. The post-mortem — 20 minutes — [10 pts]

A team's detection project went wrong in five places. For each, name the mechanism and give the fix.

(a) **[Judge]** A mixture of experts with four experts: after training, the gate sends 97% of flows to expert 1. [2 pts]

(b) **[Judge]** A representation was pretrained on 10 million unlabelled flows by predicting masked fields, then a classifier was trained on it with 2,000 labelled flows. The pretraining data included the test period. What is the learning signal of the pretraining, why can it help, when can it hurt, and is the test-period issue a problem? [2 pts]

(c) **[Judge]** Soft voting averages a calibrated logistic regression with a boosted model whose outputs are almost all 0.01 or 0.99. The ensemble behaves like the booster alone. [2 pts]

(d) **[Judge]** The team evaluated once on the test set, looked at the error per attack type, adjusted features for the worst type, and reported the new test score. [2 pts]

(e) **[Judge]** A reinforcement-learning agent manages firewall rules, rewarded by "minus the number of alerts per hour". It learns to block the traffic of the monitoring sensor. [2 pts]